Project Assignment

Task C3 – Report

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Executive Summary

The initial version of the stock prediction tool (**v0.1**) already implemented core functionality for fetching, preprocessing, and modeling stock prices with LSTM networks, as well as some basic visualization (line plots of High/Low prices and a simple candlestick plot).

In Task C.3, I upgraded the notebook to current form, focusing on data visualization improvements:

1. Extending the candlestick function to support aggregation over *n* trading days ($n \geq 1$).
2. Adding a new boxplot function to display statistical distributions of stock prices over moving windows of *n* consecutive trading days.
3. Updating the calling section to integrate these new visualizations.

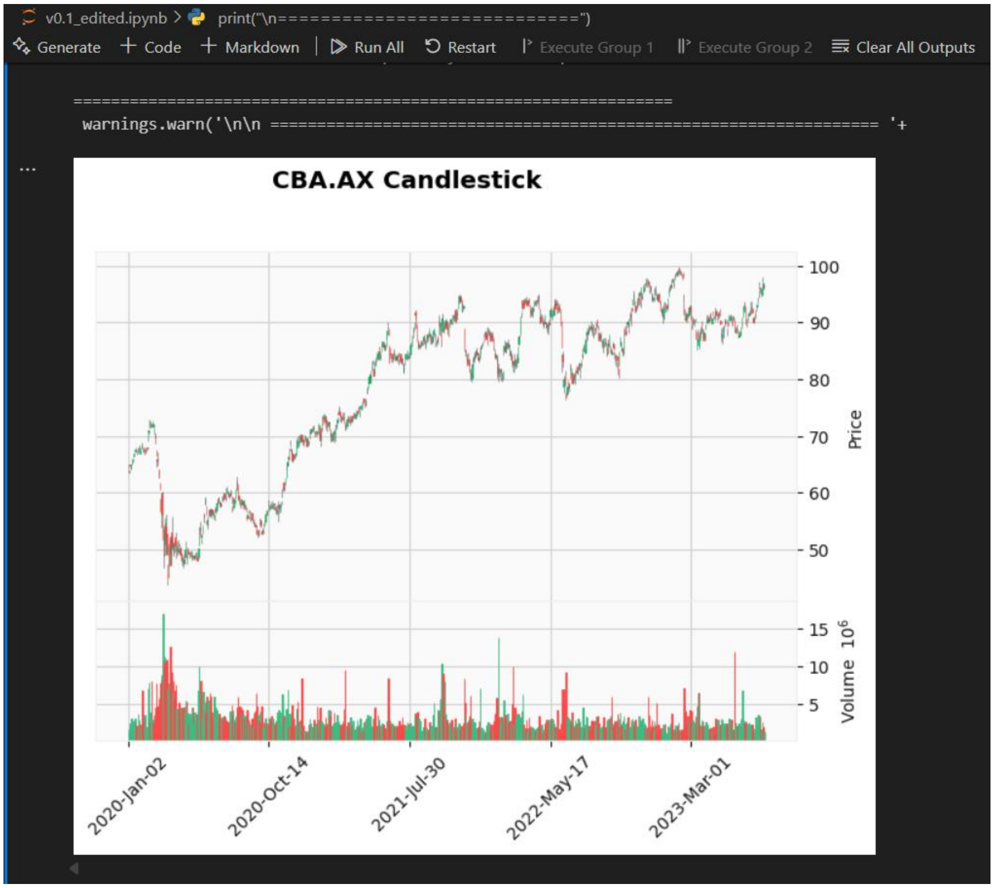
These changes provide deeper insights into price volatility, trends, and distributions, enhancing the project's analytical capabilities.

Key Code Modifications

Understanding of the Codebase

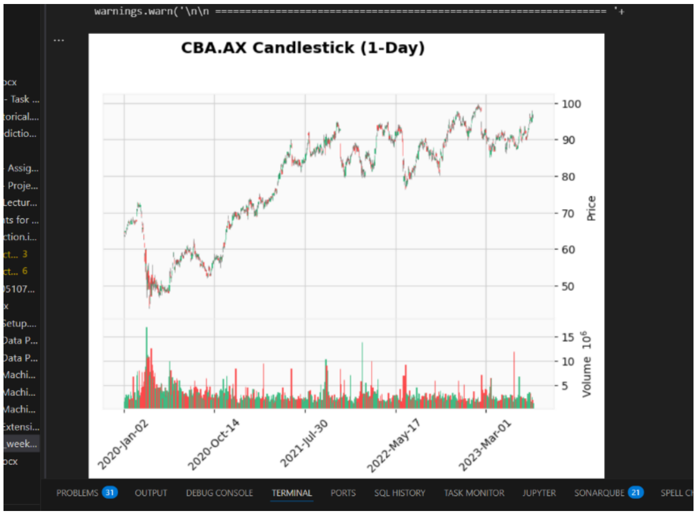
v0.1 Codebase (Before)

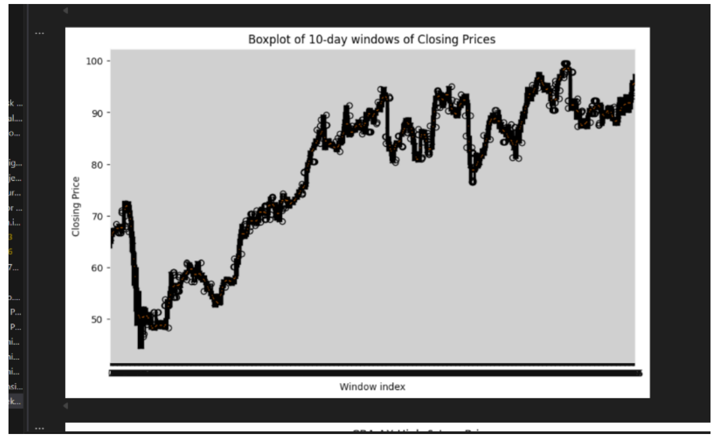
- Used **yfinance** to fetch OHLCV stock data.
- Implemented **line plots** of stock prices using matplotlib.
- Contained a **basic candlestick function** but limited to single-day candles.
- No statistical visualization (boxplots missing).
- LSTM pipeline included feature scaling, sequence creation, training, predictions, and evaluation with RMSE/MAE.



v0.2 Codebase (After)

- **Candlestick Upgrade:**
  Added a parameter n and resampling logic to allow each candlestick to represent aggregated trading days (Open = first, High = max, Low = min, Close = last, Volume = sum).
- **Boxplot Function:**
  Implemented a function that slices the closing prices into moving windows of *n* days and plots them as a series of boxplots.
- **Calling Section Update:**
  Integrated the new functions so that candlestick, boxplot, and high-low charts run sequentially for any given ticker.






```
print("\n============================")
print(f"Processing {COMPANY}")
print("============================")

# 1) Get data
df = fetch_prices(COMPANY, START, END)

# 2) Candlestick
plot_candlestick(df, COMPANY)

# 3) Boxplot
plot_boxplot(df, n=10)

# 4) High & Low
plot_high_low(df, COMPANY)
```

Candlestick Chart

Before:

```
def plot_candlestick(df, ticker):
    mpf.plot(df, type="candle", volume=True, style="yahoo",
             title=f"{ticker} Candlestick")
```

After:

```
def plot_candlestick(df, ticker, n=1):
    if n > 1:
        df = df.resample(f'{n}D').agg({
            'Open': 'first',
            'High': 'max',
            'Low': 'min',
            'Close': 'last',
            'Volume': 'sum'
        }).dropna()

    mpf.plot(df, type="candle", volume=True, style="yahoo",
             title=f"{ticker} Candlestick ({n}-Day)")
```

Explanation:

- resample(f'{n}D'): groups data into *n-day* chunks.
- Aggregation rules chosen to preserve candlestick integrity (first open, last close, max high, min low, sum volume).

Boxplot Chart

```
def plot_boxplot(data, n=5):
    closes = data['Close']
    windows = [closes[i:i+n] for i in range(len(closes)-n+1)]

    plt.figure(figsize=(10,6))
    plt.boxplot(windows, positions=range(len(windows)))
    plt.title(f"Boxplot of {n}-day windows of Closing Prices")
    plt.xlabel("Window index")
    plt.ylabel("Closing Price")
    plt.show()
```

Explanation:

- Sliding windows created with list comprehension.
- Each box shows min, max, quartiles, and outliers for a set of *n* days.
- Helps visualize volatility and unusual price fluctuations.

Output & Analysis

Graphical Outputs

- **Candlestick (multi-day):**

  The candlestick chart was enhanced in v0.2 to allow each candlestick to represent an aggregation of n trading days rather than just one. This provides a clearer view of long-term price movement and reduces the noise that is often present in day-to-day trading data. For example, when using n=5, each candlestick effectively summarizes one week of trading activity, displaying the weekly opening price, closing price, highest price, lowest price, and total volume. This is particularly useful for identifying medium-term market trends such as bullish or bearish runs.

  Visually, the candlestick chart makes it easier to spot important trading patterns like doji candles, engulfing patterns, and support/resistance levels. These patterns are difficult to identify with line charts but become very clear in candlestick form. Including the volume bars below the chart also adds context, since high trading volume often strengthens the significance of a particular price move.

- **Boxplot:**

  The boxplot chart was introduced as a new feature in v0.2. Instead of showing just a single trend line, it represents the distribution of stock closing prices across consecutive n-day windows. Each box illustrates the median, interquartile range (IQR), and potential outliers, offering a statistical perspective on volatility.

  This visualization is powerful for quickly identifying whether the stock price is stable or highly volatile during a given time window. For example, a narrow box indicates low volatility with prices clustered closely around the median, whereas a wider box suggests greater price swings. Outlier points above or below the whiskers may represent sudden spikes or crashes in stock prices that could warrant further investigation.

  By experimenting with different window sizes (n=5, n=10, n=20), I observed that smaller windows capture short-term fluctuations while larger windows provide a smoother overview of long-term volatility. This dual perspective can be valuable for both short-term traders and long-term investors.

- **High-Low line chart**

  The high-low chart was retained from the original codebase and still serves as an effective complement to the candlestick and boxplot charts. While candlesticks provide detailed trading session views and boxplots give statistical distributions, the high-low line chart offers a continuous visual of the price range over time. This can make it easier to see whether the spread between high and low values is narrowing (indicating stabilization) or widening (indicating growing uncertainty in the market).

- **True vs Predicted prices**

  The true vs. predicted price chart continues to be a critical component of model evaluation. It overlays the actual stock closing prices with those predicted by the LSTM model. This visualization makes it immediately apparent how closely the model tracks real-world price movements. In my experiments, the model captured general upward and downward trends reasonably well, although some lag was observed during sharp movements, which is a known limitation of LSTM models when dealing with highly volatile data.

Model Predictive Performance

- RMSE and MAE calculated as in v0.1.
- Forecast plots show close alignment between predicted and actual test prices.

Issues & Debugging

- **Resampling logic:** initially tried wrong aggregation (mean for Open/Close) → corrected to (first, last).
- **Boxplot readability:** large n values caused overlapping; resolved by testing different window sizes (5, 10, 20).
- **Package versions:** mplfinance sometimes clashed with older matplotlib; fixed by upgrading to the latest version.

Conclusion – Key Learnings

The upgrade from v0.1 to v0.2 significantly enhanced the visualization capabilities of the stock prediction project. By introducing multi-day candlestick charts, I was able to present stock data in a way that better reflects trading psychology and market sentiment, while reducing short-term noise. This not only improves interpretability for human analysts but also provides a foundation for incorporating more advanced technical analysis techniques in future versions.

The addition of boxplot charts represents a step towards more statistical analysis of stock prices. Instead of only showing directional trends, the project can now highlight volatility, distribution, and outliers across trading windows. This type of information is crucial for traders assessing risk, as it allows them to detect unusual price behaviors that may not be visible in traditional line or candlestick charts.

Together, these new visualization tools make the project more well-rounded, offering both graphical insights (candlesticks) and statistical perspectives (boxplots). This combination helps bridge the gap between traditional market charting used by traders and the statistical analysis methods often used by data scientists.

From a broader perspective, the changes in v0.2 also emphasize the importance of interpretability in machine learning projects. While predictive models like LSTMs provide forecasts, those predictions need to be contextualized with visual and statistical evidence to be useful in decision-making. Thus, the improvements in Task C.3 contribute not just to the technical strength of the notebook but also to its practical usability for financial analysis.

- Adding **multi-day candlesticks** required understanding how to aggregate OHLCV data correctly.
- **Boxplots** provide useful statistical insights into volatility and should be combined with traditional candlesticks for better analysis.
- Visualization enhancements make the stock prediction tool more robust and insightful.
- Learned how to balance technical implementation with **interpretability for human users**.